

Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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Introduction and Motivation

Model:

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad \begin{array}{c} \text{Diagram of three adjacent hexagons sharing a central vertex labeled } z. \text{ The top-right vertex is labeled } x \text{ and the top-left vertex is labeled } y. \end{array}$$

- Majorana fermion (parton) representation: $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = ib^\mu c$
- Parton Hamiltonian: define $ib_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij} \Rightarrow H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$
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 - Use physical **spin** operators to describe a spin system.
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- Illustrate the entanglement structure of the Kitaev model more clearly.
 - Directly solve the **entanglement spectrum (ES)**.
 - Do **NOT** use the **replica trick**.

Results

- (Reduced) Density matrix directly reflects the fermionic feature.

$$\rho = \frac{1}{2^{N/2}} \hat{\Pi} \sum_{\mathbf{EP}(\mathfrak{D})} \mathfrak{e}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}$$

$$\rho_A = \frac{1}{2^{(|\partial A|/2-1)}} \left(\frac{1}{2^{N^A/2}} \hat{\Pi}_A \sum_{\mathbf{EP}(\mathfrak{D}) \subset A} \mathfrak{e}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \right)$$

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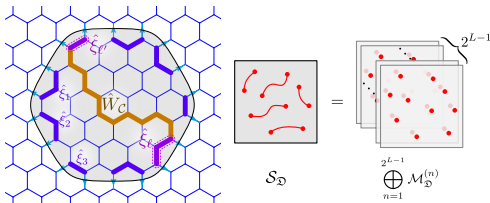
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- **Block-diagonal structure** of the DM due to the **exact 1-form Wilson sym.**:
Extensive degeneracy in the ES $\Rightarrow$ **Gauge field entanglement entropy**

$$\rho_A = \frac{1}{2^{(|\partial A|/2-1)}} \bigoplus_{n=1}^{2^{(|\partial A|/2-1)}} \varrho_A^{(n)}, \quad \ln(\text{Degeneracy}) = \left(\frac{1}{2} |\partial A| - 1 \right) \ln 2$$



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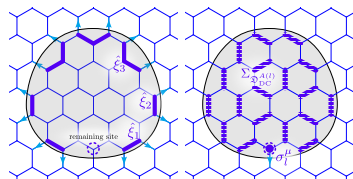
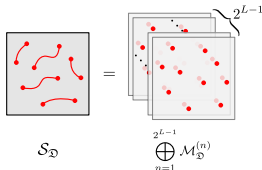
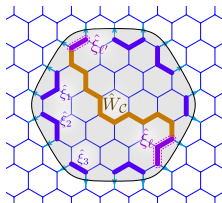
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- Subsystem fermion parity and entanglement in **odd** cases: Borrow a fermion from the gauge sector

$$\hat{\Xi}_A \equiv \sigma_l^\mu \Sigma_{\mathfrak{D}_{\text{DC}}}^{A(l)} \rightarrow \underbrace{\left(b_l^\mu \prod_{i \in A} c_i \right)}_{\text{fermion parity}}, \quad [\hat{\Xi}_A, \rho_A] = 0$$



Concluding Remarks and Outlooks

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- Expressing the density matrix in terms of the physical spin basis provides a natural starting point for **open-quantum-system**, **quantum-information**, and **quantum-resource** questions formulated in terms of **physical spin operations**.
- The stabilizer Rényi entropy calculation implies that the **magic** is governed only by the matter sector

$$M_{\alpha}(|\psi\rangle) := \frac{1}{1-\alpha} \ln \left(\sum_{S \in \tilde{\mathcal{P}}_N} \frac{1}{2^N} \text{Tr}(\rho S)^{2\alpha} \right) = \frac{1}{1-\alpha} \ln \left(\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}] \in \mathbf{G}} \mathfrak{C}_{\mathfrak{D}}^{2\alpha} \right)$$